

Right Triangle Trigonometry: SOH■CAH■TOA, Special Triangles, and Solving for Sides and Angles

Summary

This note covers the parts of a right triangle, the Pythagorean theorem, the SOH■CAH■TOA mnemonic for the six trigonometric functions, and the special integer■ratio triangles that speed up calculations. It shows how to find missing sides with the Pythagorean theorem, compute all six trig ratios, and determine unknown angles using inverse functions. Real■world examples from the lecture illustrate each step, and a flowchart summarizes the problem■solving process. The material is organized for students who missed the class and need a complete, teachable guide.

Learning Objectives

- Identify the opposite, adjacent, and hypotenuse sides of a right triangle.
- Apply the Pythagorean theorem to find missing side lengths.
- Recall and use the SOH■CAH■TOA relationships for sine, cosine, tangent and their reciprocals.
- Recognize common special right■triangle ratios (3■4■5, 5■12■13, etc.) and use them for quick calculations.
- Select the appropriate trigonometric function to solve for an unknown side or angle and use a calculator correctly.

Lecture Sections

Triangle Parts and the Pythagorean Theorem

A right triangle has one 90° angle and three distinct sides. The side opposite the right angle is the longest and is called the hypotenuse (often labeled C). The two legs that meet at the right angle are the opposite side (the leg opposite the angle of interest, usually labeled A) and the adjacent side (the leg next to the angle of interest, usually labeled B). The Pythagorean theorem states that for any right triangle, the sum of the squares of the two legs equals the square of the hypotenuse: $A^2 + B^2 = C^2$. This relationship lets you solve for any missing side when the other two are known. For example, if $A = 3$ and $B = 4$, then $C^2 = 3^2 + 4^2 = 9 + 16 = 25$, so $C = 5$. The theorem also underpins the derivation of trigonometric ratios because those ratios are simply ratios of these side lengths. Understanding which side is which is essential before applying any formula, and the hypotenuse is always the side opposite the right angle, not the side next to the angle you are examining.

Examples

Finding the hypotenuse of a 3-4-? triangle

A is 3, B is 4, apply $A^2 + B^2 = C^2 \rightarrow 9 + 16 = 25 \rightarrow C = 5$

Definitions

Opposite side: The leg of a right triangle that does not touch the angle of interest.

Adjacent side: The leg of a right triangle that shares the angle of interest.

Hypotenuse: The side opposite the right angle; the longest side of a right triangle.

Formulas

$$A^2 + B^2 = C^2$$

Instructor Emphasis

- Know the Pythagorean theorem thoroughly
- Always label the hypotenuse as the side opposite the right angle

Common Misconceptions

- Thinking the side next to the angle of interest is the hypotenuse
- Confusing which leg is opposite versus adjacent

SOH-CAH-TOA and the Six Trigonometric Functions

The mnemonic SOH-CAH-TOA helps remember the primary ratios for any acute angle θ in a right triangle. SOH stands for Sine = Opposite / Hypotenuse, CAH for Cosine = Adjacent / Hypotenuse, and TOA for Tangent = Opposite / Adjacent. The remaining three functions are reciprocals: Cosecant ($\csc \theta$) = $1 / \sin \theta$ = Hypotenuse / Opposite, Secant ($\sec \theta$) = $1 / \cos \theta$ = Hypotenuse / Adjacent, and Cotangent ($\cot \theta$) = $1 / \tan \theta$ = Adjacent / Opposite. These definitions allow you to compute any of the six functions once you know the appropriate side lengths. For instance, in a triangle where the opposite side is 4 and the hypotenuse is 5, $\sin \theta = 4/5$, $\csc \theta = 5/4$. The lecture emphasized that the ratios are always expressed with the side opposite the angle of interest, not any other angle. Mastery of these ratios is crucial for solving both side-length and angle-finding problems, and remembering the reciprocal relationships prevents common sign errors.

Examples

Computing all six ratios for a 3-4-5 triangle

$$\sin \theta = 4/5, \cos \theta = 3/5, \tan \theta = 4/3, \csc \theta = 5/4, \sec \theta = 5/3, \cot \theta = 3/4$$

Definitions

Sine (sin): Ratio of the length of the side opposite the angle to the hypotenuse.

Cosine (cos): Ratio of the length of the adjacent side to the hypotenuse.

Tangent (tan): Ratio of the opposite side to the adjacent side.

Cosecant (csc): Reciprocal of sine; hypotenuse divided by opposite side.

Secant (sec): Reciprocal of cosine; hypotenuse divided by adjacent side.

Cotangent (cot): Reciprocal of tangent; adjacent side divided by opposite side.

Formulas

$$\sin \theta = \text{opposite} / \text{hypotenuse}$$

$$\cos \theta = \text{adjacent} / \text{hypotenuse}$$

$$\tan \theta = \text{opposite} / \text{adjacent}$$

$$\csc \theta = 1 / \sin \theta = \text{hypotenuse} / \text{opposite}$$

$$\sec \theta = 1 / \cos \theta = \text{hypotenuse} / \text{adjacent}$$

$$\cot \theta = 1 / \tan \theta = \text{adjacent} / \text{opposite}$$

Instructor Emphasis

- Memorize SOH■CAH■TOA
- Remember that cosecant, secant, and cotangent are reciprocals

Common Misconceptions

- Mixing up opposite and adjacent when labeling sides
- Forgetting to flip the fraction for reciprocal functions

Special Right■Triangle Ratios

Certain integer triples satisfy the Pythagorean theorem and appear frequently in problems because they allow rapid identification of side lengths without calculation. The most common are the 3■4■5, 5■12■13, 8■15■17, and 7■24■25 triangles. Any integer multiple of these triples also forms a valid right triangle (e.g., 6■8■10 is $2 \times 3■4■5$). Recognizing these patterns saves time: if you see sides 9 and 12, you can infer the hypotenuse is 15 because $9■12■15$ is $3 \times 3■4■5$. Less common but still useful are 9■40■41 and 11■60■61. The lecture highlighted that knowing these special triplets helps you quickly find a missing side, especially in timed exams. However, you must still verify that the given numbers indeed form a Pythagorean triple; otherwise the triangle is not right■angled. Practicing with these triples builds intuition for estimating side lengths and checking work.

Examples

Identifying a 8■15■17 triangle

Given sides 8 and 17, compute missing side: $17^2 - 8^2 = 289 - 64 = 225 \rightarrow \sqrt{225} = 15$

Definitions

Pythagorean triple: Three positive integers a , b , c that satisfy $a^2 + b^2 = c^2$.

Formulas

$a^2 + b^2 = c^2$ (for integer triples)

Instructor Emphasis

- Special triplets are exam-friendly shortcuts
- Multiply a known triple by any integer to generate new right triangles

Common Misconceptions

- Assuming any three numbers form a right triangle without checking the theorem

Solving Problems: Missing Sides, Ratios, and Angles

When a problem provides two pieces of information about a right triangle, follow a systematic process. Step 1: Identify which sides are known and label them opposite, adjacent, or hypotenuse relative to the given angle. Step 2: If a side is missing, use the Pythagorean theorem ($A^2 + B^2 = C^2$) to compute it. Step 3: Choose the appropriate trigonometric ratio based on what you need—use SOH for sine, CAH for cosine, or TOA for tangent. Step 4: Substitute the known side lengths into the ratio to find the desired value. Step 5: If the problem asks for an angle, take the inverse trig function (arcsin, arccos, arctan) on a calculator set to degree mode. The lecture demonstrated this with three examples: a 3-4-5 triangle (finding all six ratios), an 8-15-17 triangle (finding missing side and ratios), and a scaled 3-4-5 triangle (15-20-25) where the missing side was inferred by scaling. Each example reinforced the same workflow, emphasizing calculator use and reduction of fractions. By mastering this flowchart, students can tackle any right-triangle question efficiently.

Examples

Finding x in a 38° angle with adjacent 42

$$\tan 38^\circ = x/42 \rightarrow x = 42 \cdot \tan 38^\circ \approx 32.8$$